Econometrics — Exam-Eve Summary (Key Points Only)

Wooldridge Ch.1–6 · Econometrics 1–2 · course R code | **[S]** not in material, standard form · **[?]** needs check · **[D]** derived from material · ✔ recomputed in code

Ch.1 · Nature of Econometrics & Data

1.1 Core definitions

- Econometrics = statistical methods to analyze economic data | mostly **nonexperimental** data
- Goals: estimate relationships | test theories | evaluate policy
- Most of econometrics = specification of the error u
- **Ceteris paribus** = "other relevant factors being equal" | causal questions hold other factors fixed
- Population = entire set to study | sample = subset | **external validity** = conclusions valid outside the study

1.2 Data types (→ method choice)

Type	Key feature
Cross-sectional	one point in time | ≈ independent (random sampling)
Time series	serially correlated | order matters | trends, seasonality
Pooled cross sections	independent samples in different years | policy before/after
Panel	**same units** over time | controls time-invariant unobservables | lags

Ch.2 · Simple Regression

Slide equations are images (not extracted) → formulas are **[S]** unless marked.

2.1 Model & OLS

$y = \beta_0 + \beta_1 x + u$ | $E(u|x) = 0$ → PRF $E(y|x) = \beta_0 + \beta_1 x$ **[S]**

$\hat{\beta}_1 = \Sigma(x_i - \bar{x})(y_i - \bar{y}) / \Sigma(x_i - \bar{x})^2$ (num = covariation, den = $SST_x > 0$)
$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$ **[S]**

- Procedure: $\bar{x}$, $\bar{y}$ → deviations → Σ cross products → Σ(x dev)² → $\hat{\beta}_1$ → $\hat{\beta}_0$
- Residual $\hat{u}_i = y_i - \hat{y}_i$ (actual − fitted) | $\Sigma\hat{u}_i = 0$ | $\Sigma x_i\hat{u}_i = 0$ | line through $(\bar{x}, \bar{y})$ **[S]**

2.2 CEO salary example (slide table)

- Fitted line from table: **salaryhat = 963.191 + 18.501·roe** **[D]**
- Slope = (2004.808 − 1072.348)/(56.3 − 5.9) = **18.501** **[D]**
- Salary in \$1,000s → CEO 12: 933 − 1459.023 = **−526.023** → over-predicted by \$526,023
- CEO 5: 1368 − 1218.508 = **+149.493** → under-predicted
- Slide typo: CEO 7 residual = 1078 − 1266.611 = **−188.611** (slide: +) ✔

2.3 Goodness of fit

$SST = \Sigma(y_i - \bar{y})^2$ | $SSE = \Sigma(\hat{y}_i - \bar{y})^2$ | $SSR = \Sigma\hat{u}_i^2$
$SST = SSE + SSR$ → $R^2 = SSE/SST = 1 - SSR/SST$ **[S]**

- Simple regression: $R^2 = corr(y, x)^2$ ✔ (regression.R)

2.4 Log forms (interpretation)

Model	β_1 means
y on x	$\Delta y = \beta_1 \Delta x$
y on log x	$\Delta y \approx (\beta_1/100)\cdot\%\Delta x$ **[S]**
log y on x	$\%\Delta y \approx (100\cdot\beta_1)\Delta x$ → **semi-elasticity**
log y on log x	$\%\Delta y \approx \beta_1\cdot\%\Delta x$ → **constant elasticity**

2.5 Assumptions & theorems

- SLR.1 linear in parameters | SLR.2 random sampling | SLR.3 sample variation in x | SLR.4 $E(u|x)=0$ | SLR.5 $Var(u|x)=\sigma^2$
- **Thm 2.1** (SLR.1–4): $E(\hat{\beta}_j) = \beta_j$ → unbiased = correct **on average** over repeated samples

Thm 2.2 (SLR.1–5): $Var(\hat{\beta}_1) = \sigma^2/SST_x$ **[S]**
Thm 2.3: $\hat{\sigma}^2 = SSR/(n-2)$ | $se(\hat{\beta}_1) = \hat{\sigma}/\sqrt{SST_x}$ **[S]**

- Var ↑ with σ^2 (noise) | Var ↓ with more variation in x

2.6 Binary x & treatment effects

- $x \in \{0,1\}$: β_0 = mean of y if x=0 | β_1 = **difference in means** **[S]**
- ATE $\tau = E[y_i(1) - y_i(0)]$ | never observe both outcomes | **random assignment** → OLS unbiased for τ
- Job training: +\$1,790 → 1,790/4,550 = **39.3%** ✔

Ch.3 · Multiple Regression: Estimation

3.1 Model & interpretation

$y = \beta_0 + \beta_1 x_1 + \ldots + \beta_k x_k + u$ **[S]** | "linear" = linear in **parameters**

- β_j = effect of x_j **holding other x fixed** (ceteris paribus)
- College GPA: hsGPA +1 (ACT fixed) → colGPA **+0.453**
- Econometrics 2: in multiple regression β_1 is the **partial slope**, not the slope of y on x_1
- Partialling out: regress x_1 on other x → residual $\hat{r}_1$ → regress y on $\hat{r}_1$ → slope = $\hat{\beta}_1$

3.2 Arrest example (implied coefficients [D])

- Prop. prior arrests +0.5 → −7.5 per 100 men → coef. **−0.150**
- Months in prison +12 → −0.408 per man → **−0.034**/month
- Quarters employed +1 → −10.4 per 100 men → **−0.104**
- Small R^2 can still give good ceteris paribus estimates

3.3 MLR assumptions

- MLR.1 linear | MLR.2 random sample | MLR.3 **no perfect collinearity** | MLR.4 $E(u|x_1...x_k)=0$ | MLR.5 homoskedasticity
- Unbiased: MLR.1–4 | **Gauss–Markov**: MLR.1–5 → OLS = **BLUE**

3.4 Omitted variable bias

True $y = \beta_0+\beta_1x_1+\beta_2x_2+u$, omit x_2:
$E(\tilde{\beta}_1) = \beta_1 + \beta_2\cdot\tilde{\delta}_1$, $\tilde{\delta}_1$ = slope of x_2 on x_1 [S]

Bias [S]	corr(x_1,x_2) > 0	corr < 0
$\beta_2 > 0$	**upward (+)**	downward (−)
$\beta_2 < 0$	downward (−)	**upward (+)**

- No bias if $\beta_2 = 0$ (irrelevant) or $\tilde{\delta}_1 = 0$ (uncorrelated)
- Ability omitted in wage eq. → return to educ biased upward [S]
- Irrelevant variable included → no bias, variance may ↑

3.5 Variance & multicollinearity

$Var(\hat{\beta}_j) = \sigma^2 / [SST_j(1 - R_j^2)]$, R_j^2 = R^2 of x_j on other x
$\hat{\sigma}^2 = SSR/(n - k - 1)$ | $se(\hat{\beta}_j) = \hat{\sigma}/\sqrt{[SST_j(1-R_j^2)]}$ | VIF = $1/(1-R_j^2)$ [S]

- σ^2 ↑ → Var ↑ | SST_j ↑ → Var ↓ | R_j^2 ↑ → Var ↑ | R_j^2 → 1 → Var → ∞
- Multicollinearity inflates only the involved variables' variances | **not** an MLR.3 violation
- Dropping a variable reduces variance but may cause OVB → bias–variance trade-off

Ch.4 · Inference

4.1 Distribution

- MLR.6: u ~ Normal(0, σ^2), independent of x | MLR.1–6 = CLM [S]
- Normality impossible: wages (≥ 0), arrests (few integers), unemployment (0/1) | log(y) may help

$t = (\hat{\beta}_j - a_j) / se(\hat{\beta}_j) \sim t_{(n-k-1)}$ (a_j = 0 usually) [S]

4.2 Critical values & rules

H_1	Reject if	Slide (5%)
$\beta_j > 0$	t > c	df 28: **1.701** ✓
$\beta_j < 0$	t < −c	df 18: **−1.734** ✓
$\beta_j \neq 0$	|t| > c	**2.06** (df ≈ 25 [?])

- Large df (two-sided): 10% **1.645** | 5% **1.96** | 1% **2.576** [S] ✓
- Steps: H_0/H_1 → t → df = n−k−1 → c (α, one/two-sided) → decide → interpret

4.3 p-value

- p = smallest significance level at which H_0 is rejected | reject ⇔ **p < α**
- Two-sided: P(|t| > 1.85) = 2 × 0.0359 = **0.0718** ✓

- Statistical ≠ economic significance → always discuss magnitude | "wrong" sign → misspecification?

4.4 Confidence interval

$\hat{\beta}_j \pm c \cdot se(\hat{\beta}_j)$, c = 97.5th pct of $t_{(n-k-1)}$ (95%) [S]

- Random bounds: covers β_j in 95% of repeated samples
- a_j outside 95% CI ⇔ reject H_0: $\beta_j = a_j$ at 5% (two-sided)

4.5 Linear combination ($\beta_1 = \beta_2$)

- $t = (\hat{\beta}_1 - \hat{\beta}_2)/se(\hat{\beta}_1 - \hat{\beta}_2)$ | se needs Cov → not in standard output [S]
- Trick: $\theta = \beta_1 - \beta_2$ → regress y on x_1 and $(x_1 + x_2)$ → se($\hat{\theta}$) directly [S]

4.6 F test

$F = [(SSR_r - SSR_{ur})/q] / [SSR_{ur}/(n-k-1)] \sim F(q, n-k-1)$
R^2 form: $F = [(R^2_{ur} - R^2_r)/q] / [(1-R^2_{ur})/(n-k-1)]$
Overall: $F = (R^2/k) / [(1-R^2)/(n-k-1)]$ [S]

- q = number of restrictions | F ≥ 0 | reject if F > c
- regression.R: (0.07449/2)/((1−0.07449)/82) = **3.30** ✓
- Restricted model with different y (e.g. rational house assessments) → use **SSR form** only [S]

Ch.5 · OLS Asymptotics

Property	Needs
Unbiased	MLR.1–4
Variance formulas, BLUE	MLR.1–5
Exact t / F	MLR.1–6
Consistent	MLR.1–4 (weaker MLR.4′ suffices)
Asymptotic t / F / CI	MLR.1–5 (no normality)

- Consistency: P(estimate close to true value) → 1 as n ↑ | minimum requirement

- MLR.4′: $E(u) = 0$, $Cov(x_j, u) = 0$ [S]
- plim $\hat{\beta}_1 = \beta_1 + Cov(x_1,u)/Var(x_1)$ | inconsistency doesn't vanish with n [S]
- Large n: t ≈ N(0,1) → t, F, CI valid **without MLR.6** | homoskedasticity still needed
- $se(\hat{\beta}_j) \approx c_j/\sqrt{n}$ → why large samples are better [S]

Ch.6 · Further Issues

6.1 Logs

- Pros: % / elasticity reading | invariant to rescaling | less outliers | helps normality, homoskedasticity
- Don't log: years | percentage points | zero or negative values

6.2 Quadratic

$y = \beta_0 + \beta_1x + \beta_2x^2 + u$ → **$\Delta\hat{y} \approx (\hat{\beta}_1 + 2\hat{\beta}_2x)\Delta x$**
Turnaround **$x^* = |\hat{\beta}_1 / (2\hat{\beta}_2)|$** [S]

- Wage peaks at **24.4 years** of experience (coefficients in image [?])

6.3 Interaction & APE

$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + u \rightarrow \partial y/\partial x_2 = \beta_2 + \beta_3 x_1$ [S]

- β_2 alone = effect at $x_1 = 0$ | center ($x_1 - \mu_1$) → coefficient = effect at the mean
- APE = average of each unit's partial effect over the sample

6.4 Adjusted R^2

$R^2 = 1 - (1 - R^2)(n - 1)/(n - k - 1)$ [S]

- regression.R: $1 - (0.92551)(84/82) =$ **0.05192** ✓
- $R^2 \uparrow \Leftrightarrow$ new regressor's $|t| > 1$ | use for nonnested models
- Never compare R^2 / $\bar{R}^2$ across **different dependent variables**
- High $R^2 \neq$ causal | low $R^2 \neq$ bad partial effects

6.5 Too many controls · prediction

- Don't control for mechanisms: beer consumption (beer tax → fatalities) | doctor visits (pesticides → health spending)
- Add regressors uncorrelated with x → $\sigma^2 \downarrow$ → more precise
- log(y) model: exp(predicted log y) underestimates → $\hat{y} = \hat{\alpha}_0 \cdot \exp(\ldots)$, $\hat{\alpha}_0 = n^{-1}\Sigma\exp(\hat{u}_i)$ [S]
- CI for mean E(y|x) narrow | prediction interval for one new y wider ($+\hat{\sigma}^2$) [S]

Ch.R · R Code — Reading the Output

R.1 regression.R (Weidmann & Deming 2021)

- n = **85** groups | y = cryptography score | x: RMET (eye), skill, Ravens | all standardized → intercept ≈ 0
- `lm(perform ~ eye)` (outcome first) | `summary()` | `confint()` | `predict()`

Model	$\hat{\beta}$ (se)	p	R^2
~ eye	0.192 (0.108)	0.079	0.037
~ skill	0.214 (0.107)	0.049	0.046
~ ravens	0.217 (0.107)	0.046	0.047
~ skill + ravens	0.171 / 0.175 (0.110 each)	0.122 / 0.115	0.074

- All recomputed in R ✓
- Overall F(2, 82) = **3.3**, **p = 0.04184** → reject H_0: both = 0
- Each significant alone (p ≈ 0.05) but not when both included (corr 0.246, computed) → partial effects, larger se
- Skill **99%** CI [−0.069, 0.497] contains 0 while p = 0.049 → consistent (not significant at 1%)
- Simple regression: $F = t^2$ ($1.997^2 = 3.989$) ✓
- `predict(skill=0.5, ravens=1.5)` = **0.3475** | (1.5, 0.5) = **0.3441** ✓
- Trap: `eye_regr` overwritten by skill model → later CI-band plot wrong

R.2 Simulation (simple regression demonstration.R)

- True $\beta_0 = 3$, $\beta_1 = 0.2$ | n = 10 | `rnorm(n, 10, 2)` → 3rd arg = **SD**
- 10,000 samples: mean $\hat{\beta}_1$ = **0.2027 ≈ 0.2** → unbiasedness ✓
- Reject 1,908 / 10,000 (19.1%) at 5% → H_0 is false → this is **power** ✓
- Larger slope or n → power ↑ | true null → reject ≈ 5% [S]
- 20 CIs (slope 2): **19/20** contain 2 ≈ 95% coverage ✓

R.3 Lab 2 & Lab 4 essentials

- $\Sigma(x - \bar{x}) = 0$ always | `var()` = SS/(**n−1**) | `sd()` = √var
- competence: SS 124.07 → var **4.772**, sd **2.185** ✓
- IQR = Q3 − Q1: competence 6.5 − 4 = **2.5** | breaks 4.5 − 1 = **3.5** ✓
- Skewed data → IQR better than SD | boxplot outlier beyond 1.5×IQR (grade 36) ✓
- `to12("12")` → "12 AM" (noon bug) | `"13" - 12` → error (character) ✓

Exam-Eve Trap Checklist

C.1 Signs & units

- Residual = **actual − fitted** ($\hat{u} < 0$ → over-predicted)
- OVB sign = sign(β_2) × sign(corr(x_1, x_2))
- log–level: **×100** for % | log–log: β = elasticity | level–log: ÷100
- Check units (CEO salary in $1,000s)
- Quadratic $\beta_2 < 0$ → maximum at x*

C.2 Degrees of freedom

- Simple $\hat{\sigma}^2$: **n − 2** | multiple: **n − k − 1** | sample variance: **n − 1**
- F: numerator q | denominator n − k − 1

C.3 Commonly confused

- R_j^2 (x_j on other x) ≠ model R^2
- Two-sided p = 2 × one tail
- One-sided vs two-sided critical value (1.701 vs 2.06)
- Multicollinearity ≠ perfect collinearity
- Unbiased ≠ consistent | BLUE needs homoskedasticity
- R^2 form of F invalid if y differs

C.4 R pitfalls

- `set.seed()` first | reused object names overwrite
- `as.integer()` before arithmetic on strings
- `==` on doubles → `round()`

C.5 Written answers

- "Holding [other x] fixed, a one-unit increase in x is associated with a $\hat{\beta}$ [unit] change in y"
- Causal claim → cite MLR.4 / random assignment
- Test: H_0/H_1 → stat → df → c or p → decision → economic meaning
- Name the assumption (SLR/MLR number) behind each property

Econometrics Midterm · Multiple Choice

Multiple Choice

1. An econometrician wants to learn the average hourly wage of all workers in Korea and collects data on 2,000 randomly selected workers. Which statement is correct?
a. A sample is used only when the population is hypothetical
b. The 2,000 workers are the population, because they are the units actually analyzed
c. The 2,000 workers are the sample, and the estimates are statements about what the population of all Korean workers is like
d. The estimates describe only the 2,000 workers and say nothing about other workers

2. Which statement best defines external validity?
a. Whether the OLS estimator is unbiased within the sample used
b. Whether the sample size is large enough to compute standard errors
c. Whether the R-squared of the regression is high
d. The validity of applying the conclusions of a study outside the context of that study

3. A researcher surveys only students at one private university in Seoul to estimate the average monthly smartphone spending of all Korean adults. What is the main concern?
a. OLS cannot be computed because the population is too large
b. The sample is unlikely to be representative of the population, so the conclusions may have weak external validity
c. The estimates are automatically unbiased for all Korean adults because the students were chosen at random within the university
d. The population becomes hypothetical, so no estimate is possible

4. Consider the model log(price) = β_0 + β_1 subsidy + β_2 log(sunlight) + u. How should β_1 be interpreted?
a. The elasticity of price with respect to subsidy
b. The slope from a simple regression of price on subsidy
c. The change in log(price) from a one-unit increase in subsidy, holding log(sunlight) fixed (a partial slope)
d. The total change in price from a one-unit increase in subsidy, including any change in sunlight it causes

5. In the model log(price) = β_0 + β_1 subsidy + β_2 log(sunlight) + u, which statement about subsidy is correct?
a. Subsidy has a linear relationship with price, but not with log(price)
b. Subsidy has a linear relationship with log(price), but not with price
c. Subsidy has a linear relationship with both price and log(price)
d. Subsidy has a linear relationship with neither price nor log(price)

6. In the model log(price) = β_0 + β_1 subsidy + β_2 log(sunlight) + u, which statement about sunlight is correct?

a. β_2 is the % change in price for a one-unit increase in sunlight

b. β_2 is the change in log(price) for a 1% increase in price

c. log(sunlight) has a linear relationship with log(price), so β_2 is approximately the % change in price for a 1% increase in sunlight, holding subsidy fixed

d. Sunlight has a linear relationship with price, so β_2 is the change in price for one more unit of sunlight

7. Suppose log(price) = β_0 + β_1 battery + β_2 battery^2 + u with $\beta_2 \neq 0$. Which statement is correct?

a. β_1 alone is the constant slope of battery for every value of battery

b. The model violates the assumption of linearity in parameters, so OLS cannot be used

c. Battery has a linear relationship with log(price) because the model is linear in parameters

d. Battery does not have a linear relationship with log(price); the effect of one more unit of battery depends on the current level of battery

8. Suppose log(price) = β_0 + β_1 battery + β_2 sunlight + β_3 battery× sunlight + u with $\beta_3 \neq 0$. Holding sunlight fixed, what is the effect on log(price) of one more unit of battery?

a. β_3· battery

b. β_1

c. β_1 + β_3

d. β_1 + β_3· sunlight

9. Which statement describes random sampling?

a. Every element has an equal probability of being assigned to the treatment or control group

b. The researcher chooses units that look typical of the population

c. Every element in the population has an equal probability of being selected into the sample

d. The sample contains more than half of the population

10. A researcher has a random sample of 500 workers and then flips a coin for each worker to decide who joins a job training program. The coin flip is an example of:

a. Random sampling

b. Perfect collinearity

c. Partialling out

d. Random assignment

11. Consider Model 2: wage = $\beta_0 + \beta_1$ educ + u, where ability is not included. People with better ability tend to get more education. Why can the OLS estimate of β_1 be biased even with a random sample?
a. There is no sample variation in educ
b. Ability goes into u and is correlated with educ, so the zero conditional mean condition E(u|educ) = 0 fails
c. Random sampling guarantees unbiasedness, so the estimate cannot be biased
d. The homoskedasticity assumption fails, which always causes bias

12. In Model 2, wage = $\beta_0 + \beta_1$ educ + u (ability omitted), suppose ability raises wage and is positively correlated with education. The OLS estimate of β_1 tends to:
a. Have a bias whose sign can never be determined
b. Understate the true return to education (downward bias)
c. Be unbiased, because ability is unobserved
d. Overstate the true return to education (upward bias)

13. The slides state: "OLS requires random sampling. It may also require random assignment." When estimating the effect of education on wages with Model 2 (wage = $\beta_0 + \beta_1$ educ + u), why would random assignment help?
a. Random assignment removes the need for any error term
b. Random assignment increases the R-squared of the regression
c. If educ were randomly assigned, it would be unrelated to ability and other factors in u, so the zero conditional mean condition would hold
d. Random assignment makes the sample representative of the population

14. A data set records the crime rate and the number of police officers for the same 150 cities in both 2020 and 2023. What type of data is this?
a. Pooled cross sections
b. Cross-sectional data
c. Panel (longitudinal) data
d. Time series data

15. A researcher draws a random sample of house prices in 1993 and a new, independent random sample of house prices in 1995 to evaluate a property tax change. What type of data is this?
a. Panel data
b. A single cross section
c. Time series data
d. Pooled cross sections

16. Which feature is typical of time series data but NOT of a cross section drawn by random sampling?
a. Observations are typically serially correlated, and their ordering conveys important information
b. Observations are more or less independent of each other
c. The ordering of observations does not matter
d. All observations are collected at the same point in time

17. In the simple regression $y = \beta_0 + \beta_1 x + u$, the OLS estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ are chosen to:
a. Minimize the sum of squared residuals
b. Make the fitted value equal to y for every observation
c. Minimize the sum of residuals
d. Minimize the sum of absolute residuals

18. In a simple regression of y on x, $R^2 = 0.25$. Which statement is correct?
a. 25% of the observations lie exactly on the regression line
b. 25% of the sample variation in y is explained by x
c. x causes 25% of the change in y
d. The estimated slope equals 0.25

19. The fitted regression is log(wage) = 0.584 + 0.083educ. How should 0.083 be interpreted?
a. A 1% increase in education raises the wage by 8.3%
b. One more year of education is associated with an approximately 8.3% higher wage
c. One more year of education raises the wage by 0.083%
d. One more year of education raises the wage by \$0.083

20. Real earnings (in \$1,000s) are regressed on train (= 1 if the person joined a randomly assigned job training program, 0 otherwise): earn = 4.55 + 1.79train. What does 1.79 measure?
a. Nothing, because OLS cannot be used when x is binary
b. Participants earn about 1.79% more than non-participants
c. The average earnings of participants are \$1,790
d. The average treatment effect: participants earn about \$1,790 more on average than non-participants

21. Under SLR.1–SLR.5, $\text{Var}(\hat{\beta}_1) = \sigma^2 / \text{SST}_x$. Which change makes $\hat{\beta}_1$ more precise (smaller variance)?
a. More sample variation in x
b. Fewer observations
c. A larger error variance σ^2
d. Replacing homoskedasticity with heteroskedasticity

22. In $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$, the OLS estimate $\hat{\beta}_1$ can also be obtained by:
a. Regressing y on x_2 first, then adding x_1
b. Regressing x_1 on x_2 (with a constant), then regressing y on the residuals from that regression
c. Regressing x_2 on the residuals from a regression of y on x_1
d. Averaging the slopes of the two simple regressions

23. Which regression violates MLR.3 (no perfect collinearity)?
a. log(salary) on log(sales) and ceoten
b. wage on educ and $educ^2$
c. wage on educ and exper, where their correlation is 0.9
d. wage on an intercept, male (= 1 if male) and female (= 1 if female)

24. The true model is $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$, but x_2 is omitted and y is regressed on x_1 only. In which case is the simple regression slope unbiased for β_1?
a. When the sample size is very large
b. When $\beta_2 = 0$ or when x_1 and x_2 are uncorrelated
c. When the R-squared is high
d. When x_2 is positively correlated with x_1 and $\beta_2 > 0$

25. Using 506 communities, the simple regression gives log(price) = 11.71 − 1.043 log(nox), while adding rooms gives log(price) = 9.23 − 0.718 log(nox) + 0.306 rooms. rooms raises price and is negatively correlated with log(nox). Which statement is correct?
a. The simple regression is biased upward (toward zero), so −1.043 understates the true effect
b. Adding rooms must always make the nox elasticity more negative
c. The simple-regression elasticity −1.043 is biased downward (too negative), as predicted by $\beta_2 > 0$ and Corr(log(nox), rooms) < 0
d. The two estimates differ only because of sampling error

26. High (but not perfect) correlation between x_1 and x_2 in a multiple regression mainly:
a. Violates MLR.3, so OLS cannot be computed
b. Increases the sampling variances of $\hat{\beta}_1$ and $\hat{\beta}_2$ but does not cause bias
c. Causes $\hat{\beta}_1$ and $\hat{\beta}_2$ to be biased
d. Inflates the variance of every coefficient, including an unrelated x_3

27. The Gauss–Markov theorem states that under MLR.1–MLR.5, the OLS estimators are:
a. Exactly normally distributed in any sample
b. The only unbiased estimators of the coefficients
c. The best estimators even when the errors are heteroskedastic
d. BLUE: they have the smallest variance among all linear unbiased estimators

28. In a regression, $\hat{\beta}_j = 0.80$ with $se(\hat{\beta}_j) = 0.10$. What is the t statistic for H_0: $\beta_j = 1$?
a. 8.0
b. −2.0
c. 2.0
d. −0.2

29. In a two-sided t test, the p-value is 0.07. Which statement is correct?
a. H_0 is rejected at the 10% level but not at the 5% level
b. H_0 is rejected at the 5% level
c. H_0 is rejected at the 1% level
d. The probability that H_0 is true is 7%

30. The 95% confidence interval for β_1 is [0.02, 0.10]. Which statement is correct?
a. In repeated samples, intervals constructed this way contain the true β_1 95% of the time, and H_0: $\beta_1 = 0$ is rejected at the 5% level (two-sided)
b. There is a 95% probability that $\hat{\beta}_1$ lies in [0.02, 0.10]
c. β_1 equals 0.06 with 95% certainty
d. H_0: $\beta_1 = 0$ cannot be rejected at the 5% level

31. To test H_0: $\beta_3 = 0$, $\beta_4 = 0$, $\beta_5 = 0$ jointly, which statement is correct?
a. The F statistic can be negative when the restrictions help the fit
b. Three separate t tests always give the same conclusion as the F test
c. The restricted model always has a smaller SSR than the unrestricted model
d. Use an F test comparing the SSR of the restricted and unrestricted models; F is large when the SSR rises a lot after imposing the restrictions

32. Which can cause the usual OLS t statistics to be invalid (not t-distributed under H_0)?
a. A sample correlation of 0.95 between two explanatory variables in the model
b. A low R-squared
c. Heteroskedasticity, or omitting an important explanatory variable
d. Having 500 observations instead of 5,000

33. For 32 chemical firms, rdintens = 0.472 + 0.321 log(sales) + 0.050 profmarg, where rdintens is R&D as a percentage of sales. If sales increase by 10%, rdintens changes by about:
a. 3.21 percentage points
b. 0.032 percentage points
c. 32.1 percent
d. 0.321 percentage points

34. Which statement about the consistency of OLS is correct?
a. Consistency requires the errors to be normally distributed (MLR.6)
b. Consistency requires homoskedasticity (MLR.5)
c. A consistent estimator is unbiased in every finite sample
d. Under MLR.1–MLR.4, the distribution of $\hat{\beta}_j$ collapses to the true β_j as $n \to \infty$, i.e. plim $\hat{\beta}_j = \beta_j$

35. pctstck = $\beta_0 + \beta_1$ funds + β_2 risktol + u satisfies MLR.1–MLR.4 (pctstck = % of pension in stocks, funds = number of mutual funds offered, risktol = risk tolerance). If funds and risktol are positively correlated and pctstck is regressed on funds only, the inconsistency in $\tilde{\beta}_1$ is:
a. Zero, because the full model satisfies MLR.1–MLR.4
b. Negative, because risktol is omitted
c. It vanishes as the sample size grows
d. Positive: plim $\tilde{\beta}_1 - \beta_1 = \beta_2\delta_1 > 0$, since $\beta_2 > 0$ and funds and risktol are positively correlated

36. cigs is the average number of cigarettes smoked per day. Can cigs have a normal distribution in the adult population, and what does this mean for OLS?
a. Yes, by the central limit theorem every variable is normal
b. No: cigs is nonnegative, piles up at zero (most adults do not smoke) and is discrete; still, with a large n, OLS t and F tests are approximately valid
c. Yes, because cigs is an average
d. No, so OLS estimates in a cigs equation are biased

37. Under MLR.1–MLR.5 but without normality (MLR.6), with a large sample:
a. t tests become invalid no matter how large n is
b. OLS becomes biased
c. The usual t and F statistics are approximately valid, because $(\hat{\beta}_j - \beta_j)/\text{se}(\hat{\beta}_j)$ is approximately standard normal
d. OLS standard errors cannot be computed

38. Under MLR.1–MLR.5, $\text{se}(\hat{\beta}_j) \approx c_j/\sqrt{\{n\}}$. If the sample size is quadrupled, the standard error is approximately:
a. Quartered
b. Halved
c. Unchanged
d. Doubled

39. Crime model (n = 2,725): narr86 is regressed on pcnv, ptime86, qemp86 (restricted model). Its residuals are regressed on pcnv, avgsen, tottime, ptime86, qemp86, giving $R^2 = 0.0015$. To test H_0: avgsen and tottime have no effect:
a. LM = $nR^2 \approx 4.09$; compared with χ^2_2 (10% critical value 4.61), H_0 is not rejected at the 10% level
b. LM = 0.0015, so H_0 is rejected at the 1% level
c. LM ≈ 4.09; compared with χ^2_5, H_0 is rejected
d. LM = $n(1 - R^2) \approx 2{,}721$, so H_0 is rejected

40. The dependent variable is changed from dollars to thousands of dollars (divided by 1,000). What happens?
a. Only the intercept changes
b. t statistics are divided by 1,000
c. R-squared falls
d. All coefficients and standard errors are divided by 1,000; t statistics and R-squared are unchanged

41. For 32 chemical firms, rdintens = 2.613 + 0.00030 sales − 0.0000000070 $sales^2$ (sales in $ millions). With salesbil = sales/1,000, the coefficients on salesbil and $salesbil^2$ become:
a. 0.00000030 and −0.0000000000070
b. 0.30 and −0.0000070
c. 0.30 and −0.0070, with the intercept and R-squared unchanged
d. 0.030 and −0.00070

42. For rdintens = 2.613 + 0.00030 sales − 0.0000000070 $sales^2$ (sales in $ millions), at what level of sales does the marginal effect of sales on rdintens become negative?
a. About 2,143
b. About 21,429, i.e. $21.4 billion
c. About 42,857
d. At sales = 0

43. A housing equation with dependent variable log(price) contains −0.545 rooms + 0.062 $rooms^2$ (other controls included). Moving from 6 to 7 rooms raises price by about:
a. 6.2%
b. 12.4%
c. −54.5%
d. 19.9%

44. In log(price) = 9.23 − 0.718 log(nox) + 0.306 rooms, what is the exact percentage change in price from one more room?
a. Exactly 30.6%
b. 0.306%
c. 3.06%
d. About 35.8%, = 100[exp(0.306) − 1]

45. In a regression where price and all explanatory variables are standardized (z-scores), the coefficient on nox is −0.340. Interpretation:
a. A 1 ppm increase in nox lowers price by 34%
b. A one standard deviation increase in nox lowers price by 0.34 standard deviation, holding the other variables fixed
c. A 1% increase in nox lowers price by 0.34%
d. nox explains 34% of the variation in price

46. log(wage) = 5.65 + 0.047educ + 0.00078educ· pareduc + 0.019exper + 0.010tenure (n = 722), where pareduc is total years of both parents' education. What is the estimated return to education when pareduc = 32?
a. 4.7%
b. 0.078%
c. About 7.2%
d. 2.5%

47. Which statement about adjusted R-squared is correct?
a. It always rises when a variable is added
b. It is always larger than R-squared
c. It can compare a model for y with a model for log(y)
d. It can fall when a variable is added; it rises only if the new variable's |t| exceeds 1

48. College GPA equation (n = 4,137): predicted colgpa = 2.70 at sat = 1200, hsperc = 30, hsize = 5, with se($\hat{y}^0$) = 0.020 and $\hat{\sigma}$ = 0.560. Which is correct?
a. The prediction interval for one student is narrower than the confidence interval for the mean
b. Both 95% intervals are 2.66 to 2.74
c. The prediction interval shrinks to zero width as n grows
d. The 95% CI for expected colgpa is about 2.66–2.74, but the 95% prediction interval for one student is about 1.60–3.80, because it also includes the error variance σ^2

49. The dependent variable is log(y). Using exp(predicted log(y)) as the prediction of y:

a. Systematically underestimates $E(y|x)$; it should be scaled up by $\hat{\alpha_0}$, e.g. $\exp(\hat{\sigma^2}/2)$ under normality or the average of $\exp(\hat{u}_i)$

b. Gives an unbiased prediction of y

c. Overestimates $E(y|x)$

d. Is valid only when R-squared is high

Econometrics Midterm · Multiple Choice — Answers

1. c
표본은 분석을 위해 선택된 모집단의 부분집합이며, 표본을 연구하는 목적은 모집단이 어떠한지 알아내는 것이다. 추정치는 표본을 바탕으로 한 '모집단에 대한' 판단이다.

2. d
외부 타당성(external validity)은 연구의 결론을 그 연구 맥락 밖에 적용하는 것이 타당한지를 말한다. 표본이 모집단을 대표하는지와 직결된다.

3. b
한 대학 학생들은 한국 성인 전체를 대표하지 못할 가능성이 크다. 표본이 모집단을 대표하는지 따져봐야 하며, 그렇지 않으면 결론을 모집단에 적용하기 어렵다(외부 타당성 약함).

4. c
다중회귀에서 β_1은 단순 '기울기'가 아니라 다른 설명변수(log(sunlight))를 고정한 상태에서의 '부분 기울기(partial slope)'이다. 종속변수가 log(price)이므로 단위도 log(price) 기준이다.

5. b
log(price)는 subsidy의 1차식이므로 선형관계이다. 그러나 price = exp($\beta_0 + \beta_1$ subsidy + ...)이므로 price와 subsidy는 선형관계가 아니다.

6. c
log-log 형태: log(sunlight)와 log(price)는 선형관계이고, β_2는 탄력성(1% 증가 → 약 β_2% 변화)이다. sunlight와 price 자체는 선형관계가 아니다.

7. d
$\Delta \log(\text{price}) / \Delta \text{battery} \approx \beta_1 + 2\beta_2$ battery 로 battery 수준에 따라 달라지므로 선형관계가 아니다. 단, 모델은 '모수에 대해' 선형이므로 OLS로 추정 가능하다.

8. d
battery와 sunlight가 상호작용하므로 battery의 효과는 $\beta_1 + \beta_3 \cdot$ sunlight 로 sunlight 수준에 따라 달라진다. 따라서 β_1만으로는 battery의 기울기가 아니며, battery와 log(price)는 선형관계가 아니다.

9. c
무작위 추출(random sampling): 모집단의 모든 원소가 표본에 뽑힐 확률이 같다. 집단 배정 확률이 같은 것은 무작위 배정(random assignment)이다.

10. d
표본을 뽑는 것은 random sampling, 뽑힌 사람을 동전 던지기로 처치/통제 집단에 나누는 것은 random assignment이다.

11. b
능력이 모델에서 빠지면 오차 u에 들어간다. 능력과 교육이 상관되어 있으면 E(u|educ)=0 (zero conditional mean)이 깨져 random sampling이어도 β_1이 편의된다.

12. d
누락변수 편의 = (능력의 효과 β_2) × (educ와 abil의 상관 방향). 둘 다 양(+)이므로 편의도 양(+), 즉 과대추정된다.

13. c
무작위 배정이면 교육 수준이 능력 등 오차 속 요인과 체계적으로 관련되지 않으므로 E(u|educ)=0이 성립하고 β_1은 불편이 된다. (대표성은 random sampling의 역할)

14. c
같은 단위(도시)를 시간에 걸쳐 반복 관측하면 패널 데이터이다. 시간불변의 관측되지 않는 특성을 통제할 수 있다.

15. d
서로 독립적으로 뽑은 두 시점의 횡단면을 합친 것이 pooled cross section이다. 같은 집을 추적하지 않으므로 패널이 아니다. 정책 변화 전후 비교에 자주 쓰인다.

16. a
시계열은 관측치 간 계열상관이 있고 순서가 중요한 정보다(추세, 계절성). 무작위 추출 횡단면은 관측치가 대체로 독립이며 순서가 무의미하다.

17. a
OLS(최소제곱법)는 잔차 제곱합 $\Sigma \hat{u}_i^2$ 를 최소화한다. 그 결과 잔차의 합은 항상 0이 된다(대수적 성질).

18. b
R^2 = SSE/SST = 1 − SSR/SST: y의 표본 변동 중 x로 설명되는 비율이다. 인과관계나 기울기 크기를 뜻하지 않는다.

19. b
log-level(semi-log) 형태: x 1단위 증가 → y 약 $100\beta_1$ % 변화. 0.083×100 = 8.3%.

20. d
x가 0/1이면 기울기 = 두 집단 평균 차이. 무작위 배정이면 OLS는 처치효과의 불편추정량이다. 비참가자 평균은 \$4,550이며, 1,790은 약 39.3% 증가에 해당한다.

21. a
분산은 오차분산 σ^2이 클수록 커지고, x의 표본변동 SST_x가 클수록 작아진다.

22. b
partialling out: x_1에서 x_2로 설명되는 부분을 제거한 잔차(x_2와 무상관인 부분)에 y를 회귀하면 그 기울기가 β_1이다.

23. d
절편이 있을 때 male + female = 1 이므로 정확한 선형관계 → 완전공선성. $educ^2$는 선형관계가 아니고, 높은 상관(0.9)은 다중공선성이지 MLR.3 위반이 아니다.

24. b
누락변수 편의 = $\beta_2 \times \delta_1$. 누락변수가 무관하거나 (β_2=0) x_1과 무상관(δ_1=0)이면 편의가 없다.

25. c
누락변수 편의의 부호 = (rooms의 효과 β_2, +) × (log(nox)와 rooms의 상관, −) = 음(−) → 하향편의. 실제로 −1.043이 −0.718보다 더 음수이다. (Wooldridge 연습문제, HPRICE2)

26. b
다중공선성은 관련된 변수의 분산만 키우고(VIF로 진단) 불편성은 유지된다. 엄밀한 의미의 MLR.3 위반이 아니다.

27. d
MLR.1–5 하에서 OLS는 선형 불편추정량 중 분산이 가장 작다(BLUE). 이분산이 있으면 더 나은 추정량이 존재한다. 정규성은 MLR.6이 필요하다.

28. b
$t = (\hat{\beta}_j - a_j)/se(\hat{\beta}_j) = (0.80 - 1)/0.10 = -2.0$. 가설값을 빼고 표준오차로 나눈다.

29. a
p-value는 H_0를 기각할 수 있는 가장 작은 유의수준이다. $p < \alpha$ 일 때만 기각: 0.07 < 0.10 이지만 0.07 > 0.05.

30. a
신뢰구간의 경계가 확률적이다(모수가 아님). 0이 95% 구간 밖에 있으므로 5% 양측검정에서 $\beta_1 = 0$을 기각한다.

31. d
$F = [(SSR_r - SSR_{ur})/q] / [SSR_{ur}/(n-k-1)]$. 제약을 가하면 SSR은 커지기만 하므로 $F \geq 0$이며, 개별 t검정과 결합검정 결론은 다를 수 있다.

32. c
이분산은 MLR.5, 중요한 변수의 누락은 MLR.4(zero conditional mean) 위반이므로 t분포가 성립하지 않는다. 설명변수 간 높은 상관(0.95)은 CLM 가정 위반이 아니다(완전공선성만 금지). (Wooldridge 연습문제 4.1)

33. b
level-log 형태: $\Delta y \approx (\beta_1/100)(\%\Delta x) = (0.321/100)\times 10 \approx 0.032$. rdintens는 퍼센트 단위이므로 '퍼센트포인트' 변화이다. (Wooldridge 연습문제, RDCHEM)

34. d
일치성: 표본이 커질수록 추정량의 분포가 참값에 몰린다. MLR.1–MLR.4면 충분하고, 더 약한 MLR.4′(E(u)=0, Cov(x_j, u)=0)만으로도 성립한다. 정규성·동분산은 필요 없다.

35. d
비일치성 = $\beta_2 \times \delta_1$, δ_1 = Cov(funds, risktol)/Var(funds). 위험선호가 클수록 주식 비중이 크므로 $\beta_2 > 0$, 상관도 양(+) → 양의 비일치성. 표본이 커져도 사라지지 않는다. (Wooldridge 연습문제 5.2)

36. b
정규분포는 연속·대칭이며 음수도 가능하다. cigs는 0에 질량이 몰려 있어 정규가 아니다. 그러나 정규성(MLR.6)은 불편성과 무관하고, 대표본에서는 점근적 정규성 덕분에 추론이 근사적으로 타당하다. (Wooldridge 연습문제 5.3)

37. c
점근적 정규성(중심극한정리): 정규성 없이도 대표본에서 t·F 검정과 신뢰구간이 근사적으로 타당하다. 단, 동분산(MLR.5)은 여전히 필요하다 — 이분산이면 대표본에서도 일반 표준오차는 틀린다.

38. b
표준오차는 1/√{n} 속도로 줄어든다. √{4} = 2 이므로 약 절반이 된다.

39. a
$LM = n \cdot R^2_u = 2{,}725 \times 0.0015 \approx 4.09$, 자유도 = 제약의 개수 q = 2, p-value ≈ 0.13. 제약 모형 잔차를 '모든' 설명변수에 회귀한다. (Wooldridge Example 5.3)

40. d
y의 단위를 바꾸면 계수와 표준오차가 같은 비율로 바뀐다. 따라서 t통계량, p-value, R^2은 변하지 않는다(통계적 결론 동일).

41. c
salesbil 계수 = 0.00030 × 1,000 = 0.30, $salesbil^2 = sales^2/1{,}000{,}000$ 이므로 계수 = $-0.0000000070 \times 10^6 = -0.0070$. 보고하기 쉬운 형태가 된다. (Wooldridge 연습문제 6.3)

42. b
전환점 $x^* = \hat{\beta}_1/(2|\hat{\beta}_2|) = 0.00030/(2 \times 0.0000000070) \approx 21{,}428.57$ ($ millions). 그 이후 한계효과가 음수가 된다. (Wooldridge 연습문제 6.3)

43. d
$\%\Delta \text{price} \approx 100[-0.545 + 2(0.062)(6)] \approx 19.9\%$. 전환점은 0.545/(2×0.062) ≈ 4.4 rooms이며, 이보다 작은 지역은 표본의 약 1%뿐이다. (Wooldridge Example 6.2, HPRICE2)

44. d
log-level의 100β 근사는 변화가 작을 때만 정확하다. 정확한 값은 $100[\exp(\beta \Delta x) - 1] = 100(1.358 - 1) \approx 35.8\%$.

45. b
표준화 계수(베타 계수)는 x가 1 표준편차 변할 때 y가 몇 표준편차 변하는지를 나타내므로 단위가 다른 변수의 효과를 비교할 수 있다. (Wooldridge Example 6.1)

46. c
$\Delta \log(wage)/\Delta educ = 0.047 + 0.00078 \times pareduc = 0.047 + 0.025 \approx 0.072$. 부모 학력이 높을수록 교육 수익률이 크다. (Wooldridge 연습문제 6.4, WAGE2)

47. d
$\bar{R}^2 = 1 - [SSR/(n-k-1)]/[SST/(n-1)]$: 자유도로 벌점을 준다. R^2은 변수를 추가하면 절대 줄지 않지만 $\bar{R}^2$은 줄 수 있다. 종속변수가 다르면(y vs log y) 비교할 수 없다.

48. d
평균의 신뢰구간: 2.70 ± 1.96(0.020). 개별값 예측구간: $se(\hat{e}^0) = \sqrt{\{0.020^2 + 0.560^2\}} \approx 0.560$, 2.70 ± 1.96(0.560). 오차분산은 n이 커져도 줄지 않는다. (Wooldridge Example 6.5–6.6, GPA2)

49. a
$E(y|x) = \exp(\sigma^2/2)\cdot\exp(x\beta)$ 이므로 $\exp(\log \hat{y})$만 쓰면 과소예측한다. $\hat{\alpha}_0 > 1$을 곱해 보정한다. (Wooldridge 6.4절, Example 6.7)

Econometrics Midterm · Short Answer

1. What is the difference between a population and a sample?

2. Why do econometricians usually study a sample instead of the population, and what is the goal of studying the sample?

3. Define external validity and state which property of the sample matters for it.

4. In log(price) = β_0 + β_1 subsidy + β_2 log(sunlight) + u, why is β_1 called a partial slope rather than simply "the slope"?

5. In log(price) = β_0 + β_1 battery + β_2 sunlight + β_3 battery×sunlight + u with $\beta_3 \neq 0$, why does battery not have a linear relationship with log(price)?

6. What is the difference between random sampling and random assignment?

7. For Model 2, wage = β_0 + β_1 educ + u (ability not included), explain why the zero conditional mean condition may be violated.

8. What is a control variable, and why is ability added in Model 1, wage = β_0 + β_1 educ + β_2 abil + u?

9. What does "ceteris paribus" mean, and why is it important in econometrics?

10. What is panel (longitudinal) data, and what is one advantage of it?

11. What does OLS minimize, and what is a residual?

12. What does R-squared measure in a regression?

13. What does the homoskedasticity assumption (SLR.5) state?

14. What does it mean that the OLS estimator $\hat{\beta}_1$ is unbiased?

15. When does omitting a relevant-looking variable x_2 NOT cause bias in the estimated coefficient on x_1?

16. What is the main consequence of multicollinearity (high but not perfect correlation among regressors) for OLS?

17. State the Gauss–Markov theorem.

18. Define the p-value of a hypothesis test.

19. Does a statistically significant coefficient always matter in practice? Answer in 1–2 sentences.

20. Interpret a 95% confidence interval for β_j.

21. Which can make the usual OLS t statistics invalid (no t distribution under H_0): (i) heteroskedasticity, (ii) a sample correlation of 0.95 between two regressors in the model, (iii) omitting an important explanatory variable? Explain briefly.

22. What does it mean that an estimator is consistent, and which assumptions are enough for OLS to be consistent?

23. pctstck = $\beta_0 + \beta_1$ funds + β_2 risktol + u satisfies MLR.1–MLR.4 (risktol = risk tolerance). If funds and risktol are positively correlated and pctstck is regressed on funds only, what is the sign of the inconsistency in $\tilde{\beta}_1$, and why?

24. Can cigs (average cigarettes smoked per day) have a normal distribution in the adult population? What does this imply for OLS inference?

25. How is the LM (n-R-squared) statistic for testing q exclusion restrictions computed, and what distribution does it follow?

26. If the dependent variable is divided by 1,000 (dollars → thousands of dollars), what happens to the OLS coefficients, standard errors, t statistics and R-squared?

27. What is a beta (standardized) coefficient, and why is it useful?

28. For rdintens = 2.613 + 0.00030 sales − 0.0000000070 sales^2 (sales in $ millions), give the turning-point formula for a quadratic and find where the marginal effect of sales becomes negative.

29. In log(wage) = $\beta_0 + \beta_1$ educ + β_2 educ· pareduc + β_3 exper + β_4 tenure + u (pareduc = both parents' education), what is the return to one more year of education, and what does β_2 measure?

30. Why is adjusted R-squared used, and when does adding a variable increase it?

31. Why is a 95% prediction interval for one individual outcome y^0 much wider than the 95% confidence interval for $E(y \mid x = x^0)$?

32. When the dependent variable is log(y), why is exp(predicted log(y)) not a good prediction of y, and how is it fixed?

Answers

1. A population is the **entire** set of individuals, objects or data points we want to study, while a sample is a **subset** of the population that is **selected** for analysis.
Keywords: entire · subset · selected

2. Studying the whole population is often too **costly** or **impossible**, so we study a sample; the goal is to find out what the **population is like** from the sample.
Keywords: costly · impossible · population is like

3. External validity is the validity of applying the **conclusions** of a study **outside** the context of that study; it depends on whether the sample is **representative** of the population.
Keywords: outside · conclusions · representative

4. β_1 is the change in log(price) from a **one-unit** increase in subsidy while **holding** the **other variables** (log(sunlight)) fixed, so it isolates the effect of subsidy alone.
Keywords: holding · other variables · one-unit

5. Because battery and sunlight **interact**, the effect of battery on log(price) is $\beta_1 + \beta_3 \cdot$sunlight, which **depends** on the level of sunlight; the slope is **not constant**.
Keywords: interact · depends · not constant

6. In random sampling every element of the population has an **equal probability** of being **selected**; in random assignment every element has an equal probability of being **assigned** to a group.
Keywords: equal probability · selected · assigned

7. Ability is omitted, so it goes into the **error** u; because people with better ability get more education, educ is **correlated** with u, so $E(u|educ) = 0$ fails and β_1 is estimated with bias.
Keywords: ability · error · correlated

8. A control variable is included to **hold** another factor **fixed** so it no longer sits in the error; adding ability helps satisfy the **zero conditional mean** condition and removes the **bias** in β_1.
Keywords: hold · zero conditional mean · bias

9. Ceteris paribus means "**other relevant factors** being **equal**"; most economic questions ask for the **causal** effect of one variable holding all other factors fixed.
Keywords: other factors · equal · causal

10. Panel data follow the **same** cross-sectional units **over time**; they can account for **time-invariant** unobservables and model lagged responses.
Keywords: same · over time · time-invariant

11. OLS chooses the estimates that minimize the **sum of squared residuals**; a residual is the **actual** value y_i minus the **fitted** value $\hat{y}_i$.
Keywords: sum of squared residuals · actual · fitted

12. $R^2 = SSE/SST$ is the **fraction** of the sample **variation** in y that is **explained** by the explanatory variable(s).
Keywords: fraction · variation · explained

13. Homoskedasticity means the **error** u has the **same** (constant) **variance** for any value of x: $Var(u|x) = \sigma^2$.
Keywords: variance · error · same

14. Unbiasedness means $E(\beta_1) = \beta_1$: if sampling and estimation were **repeated** many times, the estimates would **on average** equal the **true** population value, though one estimate may differ.
Keywords: on average · repeated · true

15. There is no omitted variable bias if the omitted variable is **irrelevant** (its coefficient is zero) or **uncorrelated** with x_1.
Keywords: irrelevant · uncorrelated

16. Multicollinearity **increases** the sampling **variance** of the affected slope estimators, but OLS remains **unbiased**; it can be detected with variance inflation factors.
Keywords: variance · increase · unbiased

17. Under **MLR.1–MLR.5**, OLS is **BLUE** (best linear unbiased estimator): it has the **smallest variance** among all linear unbiased estimators.
Keywords: BLUE · smallest variance · MLR.1

18. The p-value is the **smallest significance level** at which the null hypothesis would be **rejected**; we reject H_0 when the p-value is below the chosen level.
Keywords: smallest · significance level · reject

19. Not necessarily: statistical significance only shows the effect differs from zero, while **economic** (practical) significance depends on the **magnitude** of the coefficient.
Keywords: not necessarily · magnitude · economic

20. In **repeated** samples, intervals built this way **contain** the **true** population coefficient in 95% of cases; the bounds are random, not the parameter.
Keywords: repeated · contain · true

21. Heteroskedasticity (i) and an **omitted** important variable (iii) violate the CLM assumptions, so the t statistics are invalid; a high **correlation** of 0.95 (ii) is not a CLM violation (only perfect collinearity is ruled out).
Keywords: heteroskedasticity · omitted · correlation

22. An estimator is consistent if, as the **sample size** grows, its distribution **collapses** to the **true** parameter (plim $\tilde{\beta}_j = \beta_j$); OLS is consistent under **MLR.1–MLR.4** (even zero correlation between each x_j and u is enough).
Keywords: sample size · collapse · true · MLR.4

23. The inconsistency is **positive**: plim $\tilde{\beta}_1 - \beta_1 = \beta_2\delta_1$, with $\beta_2 > 0$ (higher **risktol** raises pctstck) and $\delta_1 > 0$ because funds and risktol are positively **correlated**; it does **not disappear** as n grows.
Keywords: positive · risktol · correlated · not disappear

24. No: most adults smoke **zero** cigarettes and cigs is **nonnegative** and discrete, so it is **not normal**; with a **large sample**, asymptotic normality still makes the t and F tests approximately valid.
Keywords: zero · nonnegative · not normal · large sample

25. Estimate the **restricted** model, regress its **residuals** on all the explanatory variables, and compute LM = $\mathbf{nR^2}$ from that auxiliary regression; under H_0 it is approximately **chi-square** with q degrees of freedom.
Keywords: restricted · residuals · nR · chi-square

26. All coefficients and standard errors are **divided by 1,000**, while the **t statistics** and **R-squared** stay **unchanged**; only the units change, not the statistical conclusions.
Keywords: divided · unchanged · t statistic · R-squared

27. A beta coefficient comes from regressing the **standardized** y on standardized x's; it is the change in y, in **standard deviations**, from a one standard deviation increase in x_j, so effects measured in different **units** can be **compared**.
Keywords: standardized · standard deviation · units · compare

28. Turning point x^* = $\mathbf{\hat{\beta}_1/(2|\hat{\beta}_2|)}$ = 0.00030/(2 × 0.0000000070) ≈ **21,428.57**, i.e. sales of about **$21.4 billion**; beyond that the estimated marginal effect of sales on rdintens is negative.

29. The return is $\beta_1 + \beta_2 \cdot$pareduc, so it **depends** on **parents' education**; the **interaction** coefficient β_2 is how much the return changes for one more year of pareduc (estimate 0.00078: at pareduc = 32 the return is about 7.2%).
Keywords: depends · parents · interaction

30. R-squared **never falls** when a variable is added, so adjusted R-squared imposes a **penalty** for extra regressors using **degrees of freedom**; adding a variable raises it only if that variable's **t statistic** exceeds one in absolute value.
Keywords: never falls · penalty · degrees of freedom · t statistic

31. The confidence interval reflects only the sampling **variance** of $\hat{y}^0$, while the prediction interval also includes the variance of the unobserved **error** for an **individual** outcome (σ^2), which does not shrink as n grows (e.g. colgpa: CI 2.66–2.74 vs PI 1.60–3.80).
Keywords: error · variance · individual

32. exp(log $\hat{y}$) systematically **underestimates** E(y|x); we **multiply** it by $\hat{\alpha}_0 = \exp(\hat{\sigma}^2/2)$ under **normal** errors, or by the average of $\exp(\hat{u}_i)$ from the residuals.
Keywords: underestimate · multiply · residuals